Thadomal Shahani Engineering College

Department of Information Technology

LABORATORY ASSIGNMENT LIST

Class: F.E. **Semester: II** **Year: 2025 -2026**
Subject: Python Programming (VSEC202) **Practical: 2Hrs /Week**

Lab Outcomes: Learner will be able to

1. Demonstrate the proficiency in basic python programming or Create and perform various operations on data structures like list, tuple dictionaries and strings.
2. Apply Control Flow and Functions for efficient coding to solve problems.
3. Demonstrate proficiency in handling file operations, managing exceptions, and developing Python Packages and executable files for modular programming.
4. Illustrate the concept of Object-Oriented Programming used in python.
5. Design Graphical User Interface (GUI) applications, utilizing appropriate Python libraries to create user-friendly interfaces.
6. Investigate and apply popular python libraries to conduct efficient data handling tasks.

Sr. No.	Name of Experiment	LO
	1.Basics of Python	
1	Write a Python program to calculate the gross salary of an employee. The program should prompt the user for the basic salary (BS) and then compute the dearness allowance (DA) as 70% of BS, the travel allowance (TA) as 30% of BS, and the house rent allowance (HRA) as 10% of BS. Finally, it should calculate the gross salary as the sum of BS, DA, TA, and HRA and display the result.	LO1
2	Write a Python program to calculate the simple interest based on user input. The program should prompt the user to enter the principal amount, the rate of interest, and the time period in years. It should then compute the simple interest using the formula Simple Interest=(Principal×Rate×Time) /100 and display the result.	
	2. Control Structure	
3	Write a Python program to print a triangle pattern (give any), emphasizing the transition from C to Python syntax.	LO2
4	. Design a Python program to compute the factorial of a given integer N.	
	3. Built in Data-Types	
5	Develop a Python program to manage a task list using lists and tuples, including adding, removing, updating, and sorting tasks.	LO1
6	Create a Python code to demonstrate the use of sets and perform set operations (union, intersection, difference) to manage student enrolments in multiple courses / appearing for multiple entrance exams like CET, JEE, NEET etc.	
7	Write a Python program to create, update, and manipulate a dictionary of student records, including their grades and attendance	
	4.File Handling	
8	Develop a Python program that reads a text file and prints words of specified lengths (e.g., three, four, five, etc.) found within the file.	LO3
9	Write a python code to take a file which contains city names on each line. Alphabetically sort the city names and write it in another file.	

5. Object oriented Programming		
10	Design a system using classes for vehicles, rental agencies, and rental transactions. Implement methods to handle vehicle availability, rental periods, pricing, and customer bookings.	LO4
6. Exception Handling		
11	Write a Python program that takes two numbers as input and performs division. Implement exception handling to manage division by zero and invalid input errors gracefully.	LO3
12	Demonstrate the use of a Python debugger.	
7. Arrays		
13	Write a Python program to create a 1D, 2D, and 3D NumPy array. Perform basic operations like reshaping, slicing, and indexing. Calculate the dot product and cross product of two vectors.	LO6
14	Develop a Python script to create two arrays of the same shape and perform element-wise addition, subtraction, multiplication, and division.	
8 GUI		
15	Develop a Python GUI application to create Student Form.	LO5
9. Pandas and Matplotlib		
16	Using the Iris Data (https://www.kaggle.com/datasets/saurabh00007/iriscsv), perform the following tasks: I. Read the first 8 rows of the dataset. ii. Display the column names of the Iris dataset. iii. Fill any missing data with the mean value of the respective column. iv. Remove rows that contain any missing values. v. Calculate and display the mean, minimum, and maximum values of the Sepal length column.	LO6
17	Using the Cars Data (https://www.kaggle.com/datasets/nameeerafatima/toyotacsv) perform the following tasks: I. Create a scatter plot between the Age and Price of the cars to illustrate how the price decreases as the age of the car increases. ii. Generate a histogram to show the frequency distribution of kilometres driven by the cars. iii. Produce a bar plot to display the distribution of cars by fuel type. iv. Create a pie chart to represent the percentage distribution of cars based on fuel types. v. Draw a box plot to visualize the distribution of car prices across different fuel types.	
10. Pattern Matching		
18	Write a Python script that prompts the user to enter E-mail and Password. Use regular expressions to validate the E-mail and Password based on these criteria: Password length should contain at least 8 characters long and maximum 12.It should have at least one uppercase letter, one lowercase letter, one digit, and one special character. E-mail should have all small letters.	LO5

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(Subject Teacher)